

10

Regression

Linear and Nonlinear Regression from the Perspective of Orthogonal Projection

Regression is a statistical method used to analyze the relationship between variables, especially to predict how one variable changes as another variable changes. Linear regression refers to a regression model in which the relationship between variables is linear, while nonlinear regression is used to handle more complex models in which the relationship between variables is not a straight line. This chapter looks at univariate linear regression, bivariate linear regression, multivariate linear regression, and polynomial regression from the perspective of orthogonal projection.

10.1 From Projection to Regression



Core skills covered in this section:

- ▶ Fitting data with a straight line represents a linear relationship between variables.
- ▶ The intercept determines the vertical position; the slope determines the degree of tilt.
- ▶ The predicted-value vector is the orthogonal projection of the target vector onto the space spanned by the explanatory variables.
- ▶ Using the Gram matrix to solve for the optimal parameters.
- ▶ The role of the generalized inverse in overdetermined systems of equations, and its use in solving for regression coefficients.
- ▶ Both the predicted values and the residuals can be expressed via the projection matrix, and they are orthogonal to each other.
- ▶ The principle of least squares: minimizing the sum of squared residuals, which can be converted into an L2-norm minimization problem.

In this section, we'll apply linear algebra tools — matrix multiplication, orthogonal projection, orthogonal matrices, orthogonal complements, the Gram matrix, rank, matrix inverses, generalized inverses, overdetermined systems of equations, and so on — to the application scenario of regression models.

What Is Linear Regression?

"Regression" refers to building a mathematical model that describes how a dependent variable changes as one or more independent variables change.

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"Linear" means the variables in the model exhibit a linear relationship, that is, the dependent variable is a weighted sum of the independent variables — a combination of addition and multiplication.

The core assumption of linear regression is that the effect of the independent variables on the dependent variable is "linearly additive," rather than exponential, multiplicative, or some more complex nonlinear relationship. This makes the model both easy to interpret and convenient to compute and analyze.

 The next section introduces polynomial regression, a common nonlinear regression model.

In this section, univariate linear regression is a mathematical method that describes the relationship between two variables using a "straight line."

The "uni" in "univariate linear regression" refers to the number of independent variables. So "univariate" means only one independent variable participates in the model — for example, predicting weight from height, where height is the "variable."

If there are two or more independent variables, it's "bivariate" or "multivariate" regression.

Imagine we have a bunch of scatter points as shown in Figure 1, and we notice they exhibit a certain "linear" relationship. We want to draw the most suitable straight line to "pass through" these points, staying as close to each point as possible, in order to predict, explain, or understand the relationship between the variables.

 Note that linear regression does not imply causation. Univariate linear regression is a commonly used statistical modeling method for describing the linear relationship between one independent variable and one dependent variable. It fits an optimal straight line so that the error between the predicted values and the actual observed values is minimized. What linear regression reveals is only the correlation between variables, not causation. Even if the model shows a linear association between the independent and dependent variables, we cannot conclude that a change in the independent variable causes a change in the dependent variable — there may be a third-party factor behind it, or it may be pure coincidence.

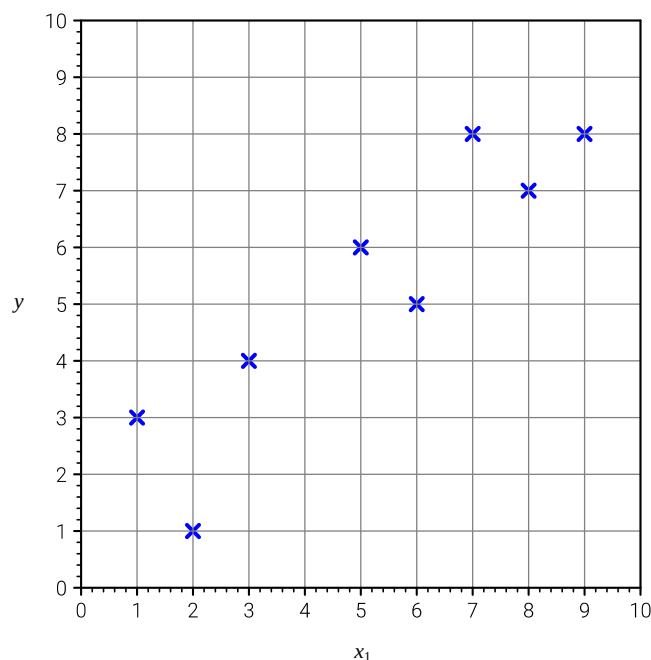


Figure 1. Scatter Data Exhibiting a "Linear" Relationship

Figure 1, the x_1 is often called the independent variable, explanatory variable, or regressor, exogenous variable, or predictor variable.

y is often called the dependent variable, explained variable, regressand, endogenous variable, response variable, and so on.

A Straight Line

Figure 2's plane, the red line can be written as

$$\hat{y} = b_0 + b_1 x_1$$

This is the slope-intercept form of a line in the plane; b_0 is the intercept, and b_1 is the slope.

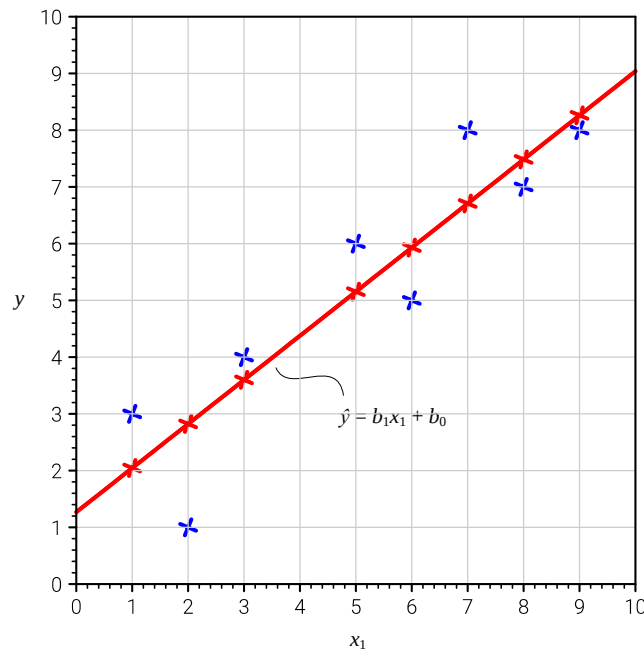


Figure 2. Univariate Linear Regression

⚠ Note that we write b_0 (the intercept) first and b_1 (the slope) second, in order to match the ordering of the parameter column vector used later in this section.

Figure 3 (a) shows the effect of the intercept on the line. The intercept b_0 determines the position where the line intersects the vertical axis (the intercept).

When the slope b_1 remains unchanged, changing the intercept b_0 shifts the entire line up or down, without changing its degree of tilt.

When the intercept b_0 is negative, the line's intersection with the vertical axis is below the origin; when the intercept b_0 is positive, the line's intersection with the vertical axis is above the origin.

Figure 3 (b) shows the effect of the slope b_1 on the line. The slope b_1 determines the degree of tilt of the line.

When the intercept b_0 remains unchanged, changing the slope b_1 rotates the line around the intercept as a pivot point.

The larger the absolute value of the slope, the steeper the line.

When the slope b_1 is zero, the line is horizontal.

When the slope b_1 is positive, the line tilts from lower-left to upper-right.

When the slope b_1 is negative, the line tilts from upper-left to lower-right.

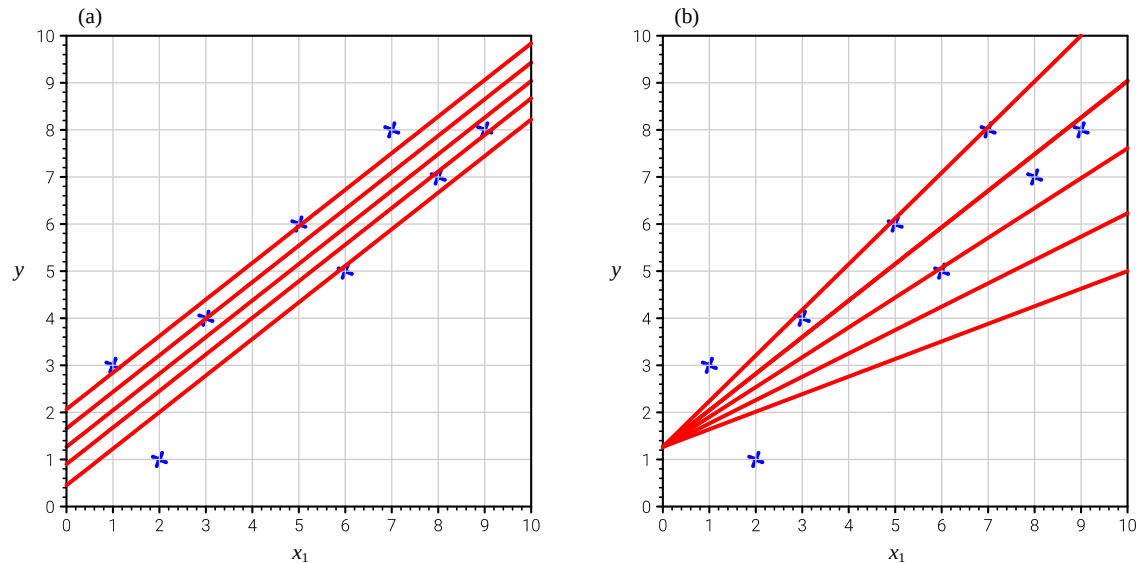


Figure 3. Effect of the Intercept and Slope on the Line

Giving variable y a "hat" $\hat{y}$ is because we reserve y for the true value — that is, the vertical-axis value of the scatter points — while $\hat{y}$ represents the "estimated" or "predicted" value, which is Figure 5's vertical-axis value corresponding to the red line.

The difference between the true y and the predicted $\hat{y}$ is exactly ε

$$\varepsilon = y - \hat{y} = y - \left(\underbrace{b_0 + b_1 x_1}_{\hat{y}} \right)$$

ε is called the residual, also known as the error term, disturbance term, or noise term.

This way, we obtain the variable relationship shown in Figure 4.

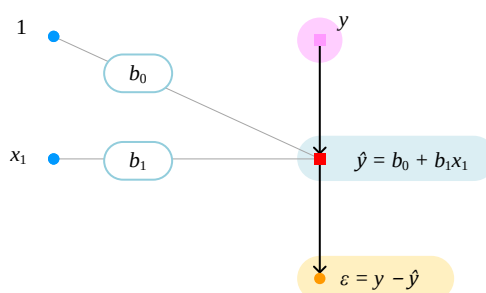


Figure 4. Variable Relationship in Univariate Linear Regression

Figure 5, the scatter point with index i (highlighted in yellow) has horizontal-axis value $x_1^{(i)}$, and vertical-axis value $y^{(i)}$; and $\hat{y}^{(i)}$ corresponds to the point on the red line $\hat{y}$, computed through the following equation

$$\hat{y}^{(i)} = b_0 + b_1 x_1^{(i)}$$

Clearly, $y^{(i)}$ and $\hat{y}^{(i)}$ do not coincide!

The difference in vertical-axis height between the two is exactly the residual

$$\varepsilon^{(i)} = y^{(i)} - \hat{y}^{(i)} = b_0 + b_1 x_1^{(i)} - \hat{y}^{(i)}$$

That is to say

$$y^{(i)} = b_0 + b_1 x_1^{(i)} + \varepsilon^{(i)}$$

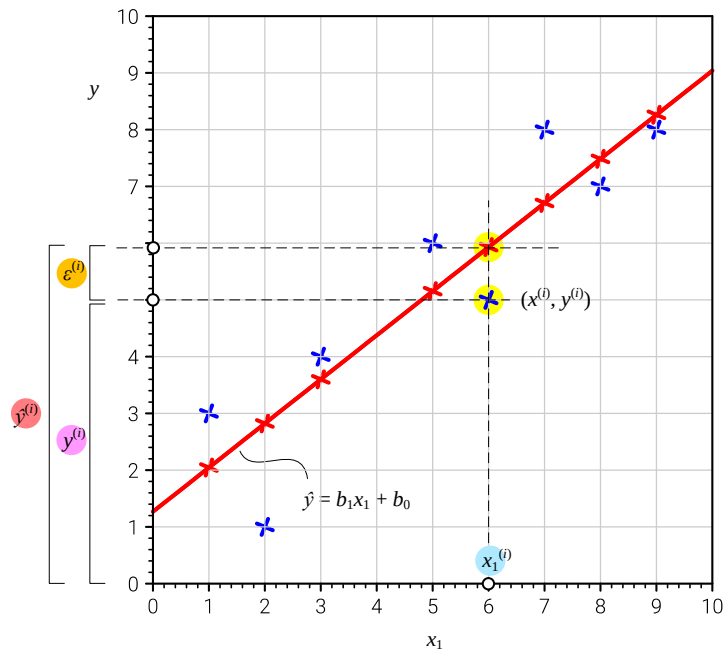


Figure 5. True Values, Estimated Values, and Error in Univariate Linear Regression

Simply put, we want to find a suitable red line such that, in Figure 6, all the residuals of the scatter points (the orange line segments) are as small as possible; this way, the red line fits the scatter points more closely.



How do we "quantify" as small as possible? This is a topic to be discussed later in this section.

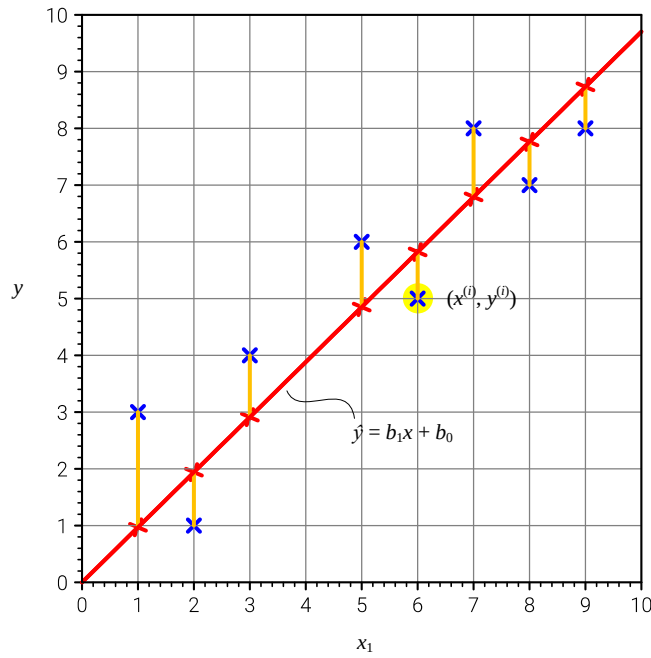


Figure 6. Error of All Scatter Points in Univariate Linear Regression (Orange Line Segments)

Written as Vectors

Writing the scatter points' horizontal coordinates as the column vector $\mathbf{x}_1$, the scatter points' vertical coordinates as the column vector $\mathbf{y}$, the estimated values represented by the red line as $\hat{\mathbf{y}}$ and the residual column vector as $\boldsymbol{\varepsilon}$, that is,

$$\mathbf{x}_1 = \begin{bmatrix} x_1^{(1)} \\ x_1^{(2)} \\ \vdots \\ x_1^{(i)} \\ \vdots \\ x_1^{(n)} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(i)} \\ \vdots \\ y^{(n)} \end{bmatrix}, \quad \hat{\mathbf{y}} = \begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \vdots \\ \hat{y}^{(i)} \\ \vdots \\ \hat{y}^{(n)} \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon^{(1)} \\ \varepsilon^{(2)} \\ \vdots \\ \varepsilon^{(i)} \\ \vdots \\ \varepsilon^{(n)} \end{bmatrix}$$

$\mathbf{x}_1$ corresponds to the horizontal-axis coordinates of the red $\times$ and blue $\times$ in Figure 6.

$\mathbf{y}$ corresponds to the vertical-axis coordinate of the blue $\times$ in Figure 6.

$\hat{\mathbf{y}}$ corresponds to the vertical-axis coordinate of the red $\times$ in Figure 6.

$\boldsymbol{\varepsilon}$ corresponds to the orange line segments in Figure 6.

Figure 1's independent-variable data column vector $\mathbf{x}$ and dependent-variable data column vector $\mathbf{y}$ have specific values

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 3 \\ 1 \\ 4 \\ 6 \\ 5 \\ 8 \\ 7 \\ 8 \end{bmatrix}$$

The Vector Addition Corresponding to the Red Line

$$\begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \vdots \\ \hat{y}^{(i)} \\ \vdots \\ \hat{y}^{(n)} \end{bmatrix} = b_0 \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ \vdots \\ 1 \end{bmatrix} + b_1 \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(i)} \\ \vdots \\ x^{(n)} \end{bmatrix}$$

$\hat{\mathbf{y}} \qquad \mathbf{1} \qquad \mathbf{x}_1$

That is

$$\hat{\mathbf{y}} = b_0 \mathbf{1} + b_1 \mathbf{x}_1$$

Here, $\mathbf{1}$ is the all-ones column vector with the same shape as $\mathbf{x}_1$, i.e., the all- $\mathbf{1}$ column vector.

 So the question becomes, how do we find suitable b_0 and b_1 ? What does this have to do with the orthogonal projection discussed earlier in this chapter?

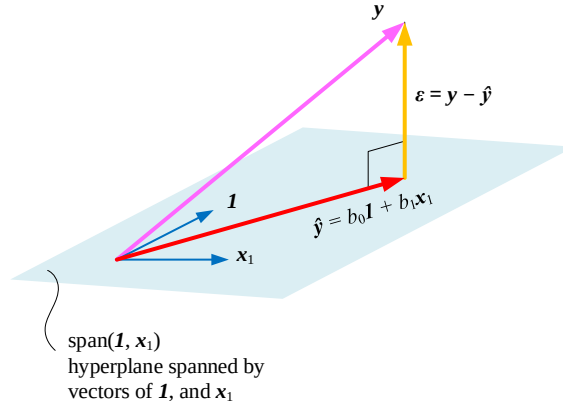
Using Orthogonal Projection to Find b_0 and b_1

The secret is hidden in the equation above!

We're very familiar with this kind of structure – it's exactly the linear combination we've repeatedly mentioned earlier in this book.

That is to say, $\hat{\mathbf{y}}$ lies in the plane spanned by $\mathbf{1}$ and $\mathbf{x}_1$, i.e., $\text{span}(\mathbf{1}, \mathbf{x}_1)$!

 Note that we assume $\mathbf{1}$ and $\mathbf{x}_1$ are linearly independent.

Figure 7. Projecting Vector y onto the Plane $\text{span}(\mathbf{1}, \mathbf{x}_1)$, Univariate Linear Regression

As Figure 7 shows, $\hat{y}$ can be written as a linear combination of $\mathbf{1}$ and $\mathbf{x}_1$, that is,

$$\hat{y} = b_0 \mathbf{1} + b_1 \mathbf{x}$$

The true-value column vector y lies outside the plane $\text{span}(\mathbf{1}, \mathbf{x}_1)$!

y , projected orthogonally onto the plane $\text{span}(\mathbf{1}, \mathbf{x}_1)$, gives $\hat{y}$. $\hat{y}$ is in $\text{span}(\mathbf{1}, \mathbf{x}_1)$; the component of y perpendicular to $\text{span}(\mathbf{1}, \mathbf{x}_1)$ is ϵ .

That is to say, y is orthogonally decomposed into two parts, namely $\hat{y}$ and ϵ .

$\hat{y}$ and ϵ sum together to construct y , that is,

$$y = \hat{y} + \epsilon = b_0 \mathbf{1} + b_1 \mathbf{x} + \epsilon$$

Rearranging the equation above, we can obtain the error vector ϵ as

$$\epsilon = y - \hat{y} = y - (b_0 \mathbf{1} + b_1 \mathbf{x})$$

The error vector ϵ is perpendicular to the plane $\text{span}(\mathbf{1}, \mathbf{x}_1)$. This is important, and we'll use it right away.

Written in Matrix Multiplication Form

As Figure 8 shows, writing the equation above in matrix multiplication form

$$\hat{y} = \underbrace{\begin{bmatrix} \mathbf{1} & \mathbf{x} \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} b_0 \\ b_1 \end{bmatrix}}_{\mathbf{b}} = \mathbf{X}\mathbf{b}$$

$\hat{y}$ corresponds to the vertical-axis coordinate of the red $\times$ in Figure 6.

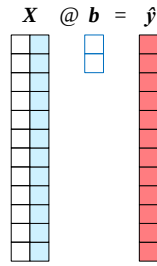
$$\mathbf{X} @ \mathbf{b} = \hat{\mathbf{y}}$$


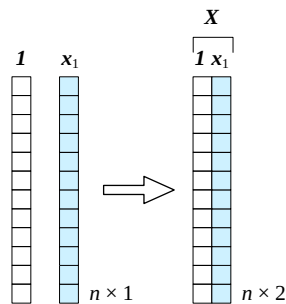
Figure 8. Matrix Operation Giving $\hat{\mathbf{y}}$, Univariate Linear Regression

We call this $\mathbf{X}$ the design matrix (design matrix). Figure 9 shows how to construct the design matrix for univariate linear regression.

The design matrix usually adds an all-ones column vector $\mathbf{1}$ as its first column to represent the intercept term, with the remaining columns holding the values of the various independent variables, making it convenient to express the regression model in matrix form.

For univariate linear regression, when the intercept is not 0, the design matrix $\mathbf{X}$ needs an all-ones column vector $\mathbf{1}$ added on its left. In Figure 1, this univariate linear regression's corresponding design matrix $\mathbf{X}$ is

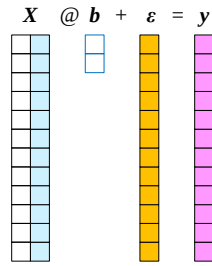
$$\mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1] = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix}$$

Figure 9. Design Matrix $\mathbf{X}$, Univariate Linear Regression

As Figure 10 shows, the true-value column vector $\mathbf{y}$ can be written as the sum of the estimated-value vector $\hat{\mathbf{y}}$ and the error vector $\boldsymbol{\varepsilon}$

$$\mathbf{y} = \hat{\mathbf{y}} + \boldsymbol{\varepsilon} = \mathbf{X}\mathbf{b} + \boldsymbol{\varepsilon}$$

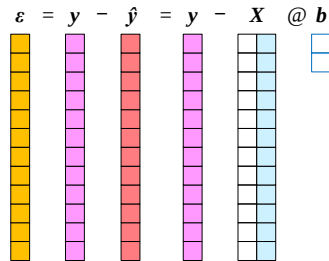
$\mathbf{y}$ corresponds to the vertical-axis coordinate of the blue $\times$ in Figure 6. $\boldsymbol{\varepsilon}$ corresponds to the orange line segment in Figure 6.

Figure 10. Matrix Operation Giving $\hat{y}$ Univariate Linear Regression

Orthogonal Relationship

As Figure 11 shows, the error vector ϵ is

$$\epsilon = y - \hat{y} = y - Xb$$

Figure 11. Matrix Operation Giving ϵ , Univariate Linear Regression

ϵ is perpendicular to X 's every column, which means,

$$\begin{cases} \mathbf{1}^T \epsilon = 0 \\ \mathbf{x}_1^T \epsilon = 0 \end{cases}$$

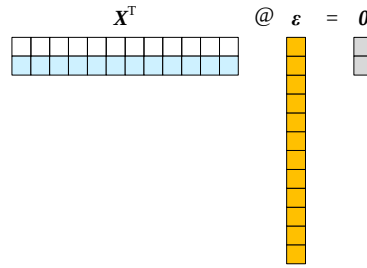
Writing the Two Equations Above as a Single Matrix Multiplication

$$\begin{bmatrix} \mathbf{1}^T \\ \mathbf{x}_1^T \end{bmatrix} \epsilon = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

That is

$$X^T \epsilon = 0$$

This point was covered in the previous chapter!

Figure 12. ϵ Perpendicular to Every Column of X , Univariate Linear Regression

Generalized Inverse

Substituting the design matrix above into the equation above gives

$$X^T (y - Xb) = 0$$

Rearranging gives the equation

$$X^T X b = X^T y$$

Earlier we assumed that $\mathbf{1}$ and $\mathbf{x}_1$ are linearly independent, so $X = [\mathbf{1}, \mathbf{x}_1]$ has full column rank. X 's Gram matrix $X^T X$ has full rank, so it is invertible.

From this we derive b 's closed-form expression

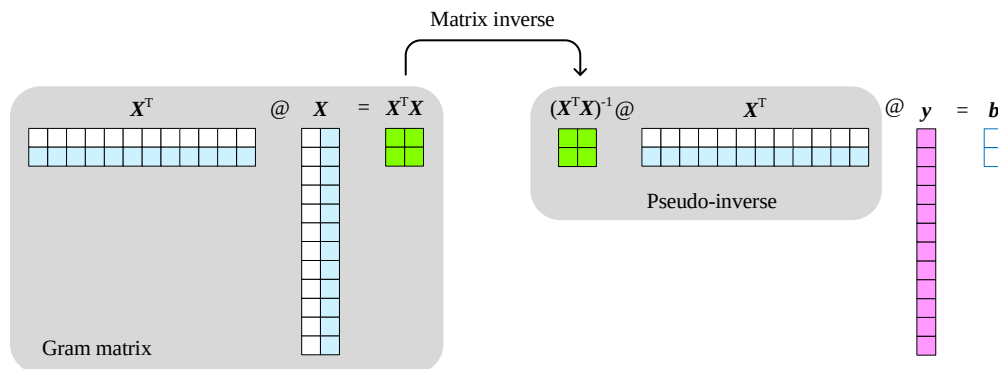
$$b = (X^T X)^{-1} X^T y$$

Figure 13 shows the workflow for computing the coefficient vector b .

? So in the space $\text{span}(\mathbf{1}, \mathbf{x}_1)$, what is b ?

According to the equation above, b is y 's projection coordinate in $\text{span}(\mathbf{1}, \mathbf{x}_1)$! b is exactly the scalar projection!

Also, as mentioned earlier in this book, $(X^T X)^{-1} X^T$ is called the pseudo-inverse, also known as the generalized inverse.

Figure 13. Computing b , Univariate Linear Regression

Substituting specific numerical values, computing the specific parameters of the red line in Figure 2

$$\begin{aligned} \mathbf{b} &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{pmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix}^T & \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix}^{-1} & \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix}^T & \begin{bmatrix} 3 \\ 1 \\ 4 \\ 6 \\ 5 \\ 8 \\ 7 \\ 8 \end{bmatrix} \end{pmatrix} \\ &= \begin{bmatrix} 8 & 41 \\ 41 & 269 \end{bmatrix}^{-1} @ \begin{bmatrix} 42 \\ 261 \end{bmatrix} = \begin{bmatrix} 0.571 & -0.087 \\ -0.087 & 0.016 \end{bmatrix} @ \begin{bmatrix} 42 \\ 261 \end{bmatrix} = \begin{bmatrix} 1.267 \\ 0.777 \end{bmatrix} \end{aligned}$$

That is to say, $b_0 = 1.267$, $b_1 = 0.777$.

⚠ Note that the computed results in this section retain only three decimal places.

Thus, the closed-form expression of the proportional function is

$$\hat{y} = b_0 + b_1 x_1 = 1.267 + 0.777 x_1$$

This way, we obtain the closed-form expression of the red line in Figure 2.

Projection Matrix

Substituting the equation above into the previous equation, we can obtain:

$$\hat{\mathbf{y}} = \mathbf{X} \underbrace{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T}_{\mathbf{b}}$$

Figure 14 shows the matrix multiplication for computing the coefficient vector $\hat{\mathbf{y}}$.

❓ Please try to analyze, in the equation above, what role $\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ plays.

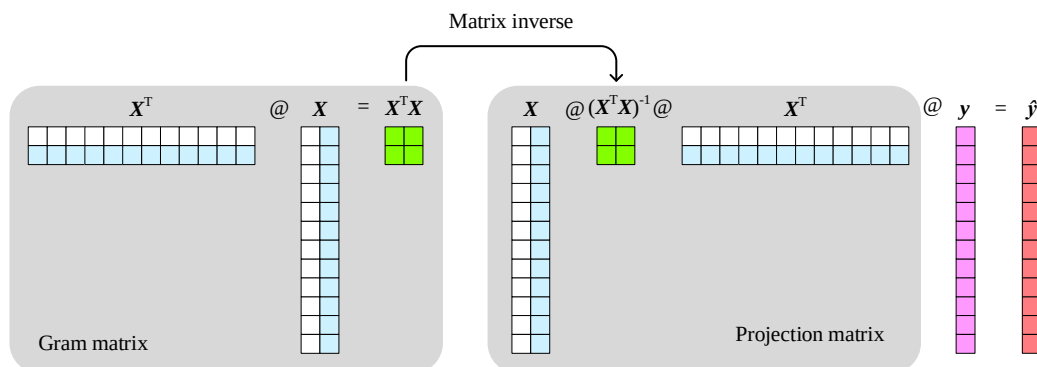


Figure 14. Computing $\hat{\mathbf{y}}$, Univariate Linear Regression

Computing the Predicted-Value Vector

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{b} = \begin{bmatrix} 1 & \mathbf{x}_1 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = b_0 + b_1 \mathbf{x}_1 = 1.267 + 0.777 \mathbf{x}_1 = 1.267 + 0.777 \times \begin{bmatrix} 1 \\ 2 \\ 3 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 2.044 \\ 2.821 \\ 3.598 \\ 5.152 \\ 5.929 \\ 6.707 \\ 7.484 \\ 8.261 \end{bmatrix}$$

$\hat{\mathbf{y}}$ corresponds to the vertical-axis coordinate of the red $\times$ in Figure 6. Clearly, all the red $\times$ marks lie on the red regression line.

Least Squares Fitting

Earlier we mentioned finding suitable b_0 and b_1 so that the residual column vector $\boldsymbol{\varepsilon}$ is as small as possible.

Recalling the content on vector norms discussed earlier in this book, the purpose of a vector norm is to compute the "size" of a vector; the most common norm is the L^2 norm. The residual column vector $\boldsymbol{\varepsilon}$'s L^2 norm is

$$\|\boldsymbol{\varepsilon}\| = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|$$

Substituting specific values, the residual column vector $\boldsymbol{\varepsilon}$ is

$$\boldsymbol{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 0.955 \\ -1.821 \\ 0.401 \\ 0.847 \\ -0.929 \\ 1.292 \\ -0.484 \\ -0.261 \end{bmatrix}$$

Each value in this column vector corresponds to the orange vertical line segment in Figure 6.

To make the computation more convenient, we square $\boldsymbol{\varepsilon}$'s L^2 norm, giving

$$\|\boldsymbol{\varepsilon}\|_2^2 = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2$$

The equation above is easily written as a matrix multiplication

$$\|\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b})$$

That is

$$\|\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} = \boldsymbol{\varepsilon} \cdot \boldsymbol{\varepsilon} = \langle \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \rangle = \begin{bmatrix} \boldsymbol{\varepsilon}^{(1)} \\ \boldsymbol{\varepsilon}^{(2)} \\ \vdots \\ \boldsymbol{\varepsilon}^{(i)} \\ \vdots \\ \boldsymbol{\varepsilon}^{(n)} \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{\varepsilon}^{(1)} \\ \boldsymbol{\varepsilon}^{(2)} \\ \vdots \\ \boldsymbol{\varepsilon}^{(i)} \\ \vdots \\ \boldsymbol{\varepsilon}^{(n)} \end{bmatrix} = (\boldsymbol{\varepsilon}^{(1)})^2 + (\boldsymbol{\varepsilon}^{(2)})^2 + \cdots + (\boldsymbol{\varepsilon}^{(i)})^2 + \cdots + (\boldsymbol{\varepsilon}^{(n)})^2$$

This is exactly least squares fitting (ordinary least squares, OLS). Simply put, least squares fitting is a mathematical method used to minimize the sum of squared errors between the predicted values and the true values when fitting a model.

Substituting specific values, ϵ 's L^2 norm squared is

$$\|\epsilon\|_2^2 = \epsilon^T \epsilon = \begin{bmatrix} 0.955 \\ -1.821 \\ 0.401 \\ 0.847 \\ -0.929 \\ 1.292 \\ -0.484 \\ -0.261 \end{bmatrix}^T \begin{bmatrix} 0.955 \\ -1.821 \\ 0.401 \\ 0.847 \\ -0.929 \\ 1.292 \\ -0.484 \\ -0.261 \end{bmatrix} = 7.949$$

As Figure 15 shows, the side length of each square in the figure is exactly a residual. The scalar value of the equation above is the sum of the areas of the squares in the figure.

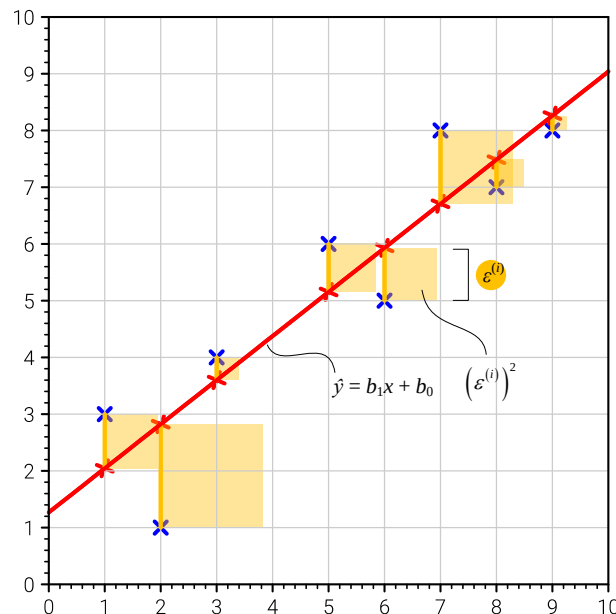


Figure 15. The Sum of Squared Residuals Corresponds to the Sum of the Square Areas in the Figure

? How do we solve for the minimum of the expression above? This requires optimization tools, a topic to be discussed in Advanced Math Is Not Hard, part of the "Math Is Not Hard" series.

Obtaining a red line is not the endpoint of linear regression analysis – in fact, it's just the beginning. We also need various probability and statistics tools to assess the reliability of this line, such as evaluating whether the model is significant, whether the variables are significant, and whether the residuals satisfy assumptions such as normality and homoscedasticity. At the same time, we need to estimate the uncertainty of the parameters, compute confidence intervals, test the model's goodness of fit, and even investigate possible multicollinearity among the variables. Only after fully understanding these statistical characteristics can we make a truly scientific judgment about the linear model's explanatory power, predictive power, and its prospects for application to real-world problems.



Linear regression analysis is a topic to be discussed in Probability and Statistics Is Not Hard, part of the "Math Is Not Hard" series.

Code Implementation

Code 1 shows how to solve univariate linear regression using Python code.

- a** Creates two column vectors x and y : x is the input value, i.e., the "independent variable," and y is the corresponding output value, i.e., the "dependent variable." `.reshape(-1,1)` means converting the original one-dimensional array into a two-dimensional "column vector," because it must be in two-dimensional form for the matrix operations that follow.
- b** Constructs a "design matrix" X . `np.ones_like(x)` creates an all-ones column vector the same size as x , serving as the column for the constant term (also called the intercept term); `np.column_stack` concatenates this all-ones column with x , and the resulting X is the input matrix we'll use for regression — each row corresponds to a sample, and each column corresponds to a feature, with the first column being the constant 1 and the second column being the original x values.
- c** Is the most critical line: it computes the linear regression's parameter vector b , that is, the intercept and slope of the regression line. This uses the standard formula for the least squares method; the overall meaning is: using a matrix formula to "fit" the relationship between the input X and the output y , obtaining the most reasonable line.
- d** Computes the predicted y values on x_array , that is, the position of the regression line at new data points, used to draw a smooth fitted curve.
- e** Makes predictions on the training data — that is, using the training data x as input, we obtain the predicted values y_pred , with the goal of comparing them against the true values y to see how well the fit performs.
- f** Computes the prediction error, also called the residual. We subtract the predicted value y_pred from the true y , and the result, error, is the "error vector" for each data point, reflecting the deviation between the predicted and actual values.
- g** Computes the sum of squares of the error vector, called the square of the L2 norm, also known as the "residual sum of squares." It's a metric for evaluating the quality of the fit — the smaller it is, the more closely the regression line fits the data points.
- ?** `LA_10_01_01.ipynb` also has a section of visualization code; the code is straightforward, so please study it on your own.

Code 1. Univariate Linear Regression |  `LA_10_01_01.ipynb`

```

## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 数据
a x = np.array([1,2,3,5,6,7,8,9]).reshape(-1,1) # 自变量 x
  y = np.array([3,1,4,6,5,8,7,8]).reshape(-1,1) # 因变量 y

# 构造设计矩阵, 增加全1列向量
b X = np.column_stack((np.ones_like(x), x))

## 计算参数
c b = np.linalg.inv(X.T @ X) @ X.T @ y

# 格拉姆矩阵
X.T @ X

# 格拉姆矩阵的逆
np.linalg.inv(X.T @ X)

## 计算预测值
x_array = np.linspace(0,10)
X_array = np.column_stack((np.ones_like(x_array), x_array))

d y_array_pred = X_array @ b
e y_pred = X @ b

## 误差向量
f error = y - y_pred

# 误差向量的L2范数的平方
g error.T @ error

```



Please study LA_10_01_02.ipynb on your own. This code uses the LinearRegression tool provided by Scikit-learn, which automatically completes the model training and prediction steps; the underlying mathematical principle is exactly the same as the least squares method, and the results are also exactly the same as LA_10_01_01.ipynb.

Proportional Function: Without an Intercept Term

To close this section, let's look at a special univariate linear regression model — a univariate linear model that passes through the origin. If the linear regression model adopts a proportional function, that is, the intercept b_0 is 0, then we only need to estimate b_1 , and the resulting line passes through the origin.

Since we want to fit the data with a proportional function, the intercept b_0 is set to 0; thus, the design matrix $\mathbf{X} = \mathbf{x}_1$, and $\mathbf{X}$ does not need the all-ones column vector $\mathbf{1}$ added on its left. This makes it easy to compute the parameter b_1

$$b_1 = (\mathbf{x}_1^T \mathbf{x}_1)^{-1} \mathbf{x}_1^T \mathbf{y} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{pmatrix}^T @ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{pmatrix} @ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{pmatrix}^T @ \begin{pmatrix} 3 \\ 1 \\ 4 \\ 6 \\ 5 \\ 8 \\ 7 \\ 8 \end{pmatrix} = 269^{-1} \times 261 = 0.970$$

Thus, the closed-form expression of the linear regression model passing through the origin is

$$\hat{y} = b_1 x_1 = 0.970 x_1$$

The equation above corresponds to the red line in Figure 16.

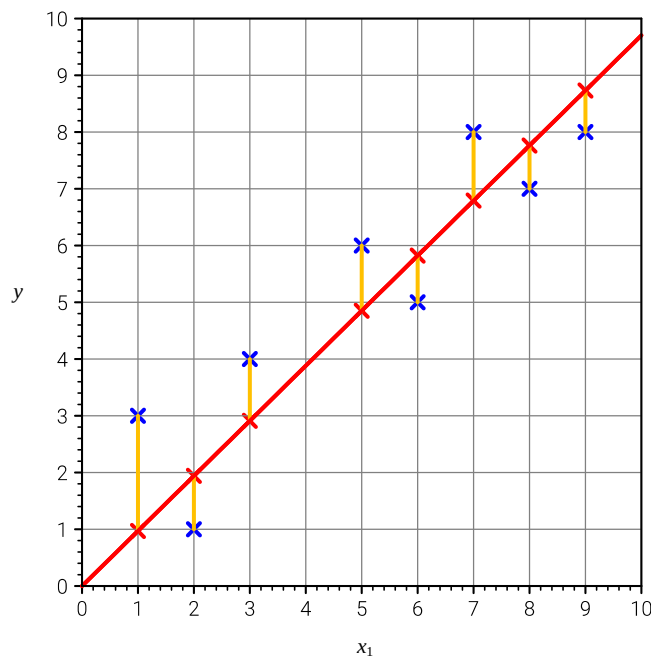


Figure 16. Proportional Function Model

Computing the Predicted-Value Vector

$$\hat{\mathbf{y}} = b_1 \mathbf{x}_1 = 0.970 \times \begin{bmatrix} 1 \\ 2 \\ 3 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 0.970 \\ 1.941 \\ 2.911 \\ 4.851 \\ 5.822 \\ 6.792 \\ 7.762 \\ 8.732 \end{bmatrix}$$

Computing the Error Vector

$$\boldsymbol{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 2.029 \\ -0.940 \\ 1.089 \\ 1.148 \\ -0.821 \\ 1.208 \\ -0.762 \\ -0.732 \end{bmatrix}$$

Substituting Specific Values to Compute the Sum of Squared Errors

$$\|\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} = \begin{bmatrix} 2.029 \\ -0.940 \\ 1.089 \\ 1.148 \\ -0.821 \\ 1.208 \\ -0.762 \\ -0.732 \end{bmatrix}^T \begin{bmatrix} 2.029 \\ -0.940 \\ 1.089 \\ 1.148 \\ -0.821 \\ 1.208 \\ -0.762 \\ -0.732 \end{bmatrix} = 10.762$$



Please study LA_10_01_03.ipynb on your own. This code solves the univariate linear model above with intercept 0 (the proportional function).



Please use tools such as DeepSeek/ChatGPT to complete the following exercises in this section.

Q1. Please solve by hand, and write code to solve the following overdetermined system of equations.

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 4 \\ 3 \end{bmatrix}$$

Q2. The following code generates sample data for the independent and dependent variables; please complete the code to construct a univariate regression model.

```
import numpy as np
np.random.seed(0)
x = np.random.rand(50) * 10
true_slope = 0.8
true_intercept = 2
y = true_slope * x + true_intercept + noise
```